

IC Lab Final

2018 Spring

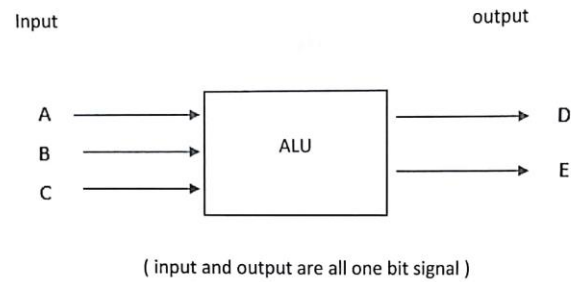
Name:

Student ID:

Total score: 100%

1. [15%] Please answer the following questions:
 - a. [3%] Please explain CDC.
 - b. [8%] In order to solve CDC problem, you need to use Synchronizer with XOR. Please draw the "2-stage Synchronizer with XOR" with Flip-Flop and XOR.
 - c. [4%] And if we use 1-stage Synchronizer with XOR, what problem may occur?
2. [3%] Describe 3 different Voltage Scaling Strategies (SVS, MVS, DVFS).
3. [2%] What problems might occur when 2 power domains at different voltage levels are connected together (list one)? And how to solve it?
4. [10%] In Dynamic Clock Gating Technique, And-gating and Or-gating both suffer from glitch problem. Please draw the glitch-free Dynamic Clock Gating block diagram with input signals: clock, awake(sleep) and output signal: G_clk_and(G_clk_or) [6%] and their waveform (please also draw glitch in awake(sleep) signal and explain how the problem can be avoided? [4%]
5. [10%] Please answer the following questions:
 - a. [5%] Please briefly describe IR drop.
 - b. [5%] Please briefly describe the way to prevent IR drop.
6. [6%] Why we need library characterization? (one reason)
7. [10%] Please explain the following:
 - a. [4%] What is the usage of Halo in floorplan
 - b. [3%] What is Metal migration
 - c. [3%] What is Antenna effect
8. [6%] List three of Signal Integrity (SI) Issue
9. [4%] List two routing method to prevent Signal Integrity (SI) Issue

10. [7%] Please answer this question:



```
module TESTBED;
    reg clock ;
    INF inf(systemclock);
    PATTERN test(.clk(clock) , .inf(inf.PATTERN));
    ALU dut(.clk(clock) , .inf(inf.DESIGN));
endmodule
```

Above figure is the I/O port for the ALU design and the module TESTBED .

You need to declare and finish the interface content used in the TESTBED module .

```
Interface .....;
.
.
.
endinterface
```

11. [8%] Please answer the following questions:

```
covergroup test @(posedge clk);
    option.per_instance = 1;
    option.at_least = 100;
    coverpoint inf.signal_0{
        bins signal_0 [] = ([0:4] => [0:4]);
    }
    signal1:coverpoint inf.signal_1{
        bins signal_1 = 1 ;
    }
    coverpoint inf.signal_2{
        option.auto_bin_max = 16;
    }
    signal3:coverpoint inf.signal_3{
        bins s3_0 = 0;
        bins s3_1 = 1;
        bins s3_2 = 2;
    }

    cross signal1 , signal3 ;
endgroup
```

Above is a covergroup declaration.

- a. [4%] If signal_2 is a 10 bits signal, how many numbers will contain in one bin?
- b. [4%] In what condition that this covergroup will achieve 100% coverage?
(you need to write down each condition clearly)

1-23
63
980

12. [9%] Nondeterministic constant is very useful during formal testing.
Please use this concept to declare a 4 bits signal DVAL and write an assertion to check if data comes in the fifo, eventually must come out .

(the input of fifo is data_in and output of fifo is data_out , they are also 4 bits)

Assertion1: assertion property (@(posedge clk)) ;

13. [10%] There are different kind of coverage report in the jaspergold .
Please specify the concept of COI and proof coverage ?

What is the difference between them ?

If we want to get the COI or proof coverage report , do we need to prove all the assertions ?